



V93000 Specifications

Pin Scale SL Digital Card

Covers Part Number E8080B, E8080C and E8080E.

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Pin Scale SL Digital Card

Note

The specifications are subject to change without notice. Make sure to refer always to the latest revision found in the most recent SmarTest documentation (Reference > Technical specifications).

General

Pin Scale SL is a highly integrated digital card for the V93000 platform to address high speed serial link needs in engineering and production. Pin Scale SL can switch between high speed lane pin electronic and low speed universal I/O digital pin electronic within test program execution.

This digital card addresses very high speed serial port testing up to 16 Gbps as well as structural testing over broadside access and functional I/O testing to 1600 Mbps in a single instrument. In high speed mode the instrument serves complete lanes configured with dedicated Tx and Rx pin pairs. In lower speed mode the digital card features a universal pin electronic implementation, which supports single ended I/O, differential input and differential output operation or any arbitrary mix of operating modes. All resources can be configured for high speed and lower speed operation on a per lane granularity, which maximizes the utilization of the digital card across a wide range of applications. Pin Scale SL is plug in compatible and mixable with existing Pin Scale digital cards (PS1600 and PS9G), as well as with the entire fleet of DC, mixed signal and RF options available for the V93000 platform. Pin Scale SL is built upon the proven V93000 test processor per pin architecture, features true per pin AC and DC resources, sequencer per pin structure, fail capture per pin, integrated TMU per pin plus new high speed backbone communication technology to enable unprecedented data transfer from and to the host computer.

High Speed Serial Port Test Support

Pin Scale SL features dedicated capabilities to support high speed serial PHY characterization as well as production test. In either operational mode stimulus and response side feature AC and DC parametric capability to stress and characterize key parameters such as output jitter, input sensitivity, jitter tolerance, TX transition times as well as RX sensitivity. The flexibility of the high speed pin electronic support a wide DC range and selectable AC or DC coupled connection to the device. Per Pin Timing generation (local synthesis) supports frequency offset margin and screening tests.

Further information

For more information about the Advantest V93000 SOC Series, please visit www.advantest.com.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

PSSL overview

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board, *valid system calibration* and a fixture delay measurement (TDR), these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Test Processor Per-Pin architecture

Maximum channel count	8 differential lanes / 32 channels per card
Parallel vector memory ^[1]	16 MByte (64 MVectors at x4 mode) standard, 32 MByte (128 MVectors at x4 mode) optional, 64 MByte (256 MVectors at x4 mode) optional, 112 MByte (448 MVectors at x4 mode) optional, 224 MByte (896 MVectors at x4 mode) optional,
Maximum scan depth ^[2]	28.5 GVectors (at x4mode)

High Speed I+O Differential Lane (Precision timing operation mode)

	E8080B	E8080C	E8080E
Maximum data rate	13.3 Gbps (x12 mode, $V_{swing} \leq 500$ mV) ^[3] 15.5 Gbps (x14 mode, $V_{swing} \leq 500$ mV) ^[3] 16 Gbps (x16 mode, $V_{swing} \leq 500$ mV) ^[3]	13.3 Gbps (x12 mode, $V_{swing} \leq 500$ mV) ^[3] 15.1 ^[4] Gbps (x14 mode, $V_{swing} \leq 500$ mV) ^[3] 15.1 ^[4] Gbps (x16 mode, $V_{swing} \leq 500$ mV) ^[3] 16 Gbps (x16 mode, clock signal only, $V_{swing} \leq 500$ mV) ^[3]	13.3 Gbps (x12 mode, $V_{swing} \leq 500$ mV) ^[3] 15.5 ^[4] Gbps (x14 mode, $V_{swing} \leq 500$ mV) ^[3] 16.0 ^[4] Gbps (x16 mode, $V_{swing} \leq 500$ mV) ^[3]
Maximum clock rate	6.7 GHz (x12 mode) 7.8 GHz (x14 mode) 8 GHz (x16 mode)		

^[1] Vector memory is used for pattern, setup data and result capture

^[2] Maximum scan depth Support total of 8 chains per instrument, 3.584GVector max (drive only, single bit mode) or 1.792 GVectors in x4 mode for EACH chain. Total data volume per instrument ~ 28.5 GVectors (in x4 mode).

^[3] V_{swing} specified into 50 ohm

^[4] This specification is valid for a temperature of the cooling unit for the system set to a value between 25°C and 30°C for the secondary water temperature. For E8080E limited up to 25°C. For details on the implementation contact your local sales representative.

Low Speed Full IO Single Ended on all pins (Pin Scale 1600 compatible operation mode)

Maximum data rate	1600 Mbps (x4mode, $V_{\text{swing}} \leq 1.8 \text{ V}$ ^[3]
	1200 Mbps (x4mode, $V_{\text{swing}} \leq 3.0 \text{ V}$ ^[3]

Test programming and debugging

The SmarTest production test environment includes manufacturing test interfaces to operators, handling equipment, and factory networks. A graphical testflow editor links device tests into a production-ready test program, where the tests are set up via fill-in-the-blank test methods. User-specific tests are programmed with test methods in Java. Links are available for design-to-test conversion. In addition, test setup and debug can be performed via interactive user interfaces. Result analysis tools are available for error locations, timing behavior, analog waveforms, shmoo plots and pin margin tests.

Compatibility

This digital card is supported starting with SmarTest 8.1.0. It will mix and match with existing Pin Scale (PS1600 and PS9G), DC Scale and Wave Scale (WSMX and WSRF) capabilities. Configuration recommendations are available to optimize fleet utilization. PSSSL has two operation modes. Pin Scale 1600 compatible operation mode and Precision Timed mode. Precision Timed mode enables high speed testing capabilities.

Calibration and operation

Warm-up time	60 minutes
Basic maintenance period	6 months
Base calibration period (traceable calibration)	6 months
Calibration period ^[5]	3 months

Environmental

Operating	15°C to 30°C (59°F to 86°F) ^[6]
Specification warranted temperature	20°C to 30°C (68°F to 86°F) ^[6]
Maximum humidity at 30°C	< 70% R. H., non-condensing
Storage (without water)	-40°C to +70°C

Maximum External Input Voltage

Maximum range for external voltage applied to pin ^[7] - 6.0V to +9.0V

^[5] Valid at ambient temperature within $\pm 2.5 \text{ K}$ of calibration temperature. For temperature requirements during calibration please see "Maintenance Guide".

^[6] For E8080E limited up to 25°C.

^[7] valid for DC connection; for AC connection -1V to 4V (50 ohm termination)

PSSL Precision Timed

OTA/EPA

Test method: TX to RX loop using PRBS 2¹⁵ pattern and verified with $V_{\text{swing}} = 100\text{mV}$ and 500mV into 50 ohm, loop includes standard high speed DUT board traces.

For more details, see **DUT board design reference > Electrical design: General rules > High-speed DUT board design (91603)**.

OTA/EPA Differential

	8 Gbps per lane	8 Gbps board 8 lanes	E8080B/E8080E: 16 Gbps per lane E8080C: 15.1 Gbps per lane	E8080B/E8080E: 16 Gbps board 8 lanes E8080C: 15.1 Gbps board 8 lanes
Overall Timing Accuracy (t_{OTA})	< ± 12 ps	< ± 28 ps	< ± 14 ps	< ± 36 ps
Edge Placement Accuracy ($t_{\text{EPA}} = t_{\text{OTA}}/2$)	< ± 6 ps	< ± 14 ps	< ± 7 ps	< ± 18 ps

OTA/EPA Single Ended

	8 Gbps per pin pair	8 Gbps board 32 pins	E8080B/E8080E: 16 Gbps per pin pair E8080C: 15.1 Gbps per pin pair	E8080B/E8080E: 16 Gbps board 32 pins E8080C: 15.1 Gbps board 32 pins
Overall Timing Accuracy (t_{OTA})	< ± 14 ps	< ± 36 ps	< ± 18 ps	< ± 38 ps
Edge Placement Accuracy ($t_{\text{EPA}} = t_{\text{OTA}}/2$)	< ± 7 ps	< ± 18 ps	< ± 9 ps	< ± 19 ps

For OTA/EPA Single Ended only one connection at once of the differential receiver can be selected for single ended comparison and 500mV are applied. Typical same value applies for 100mV.

Edge placement

Edge placement resolution	1 ps
Edge placement range ^[8]	- 4 to 32 sequencer periods

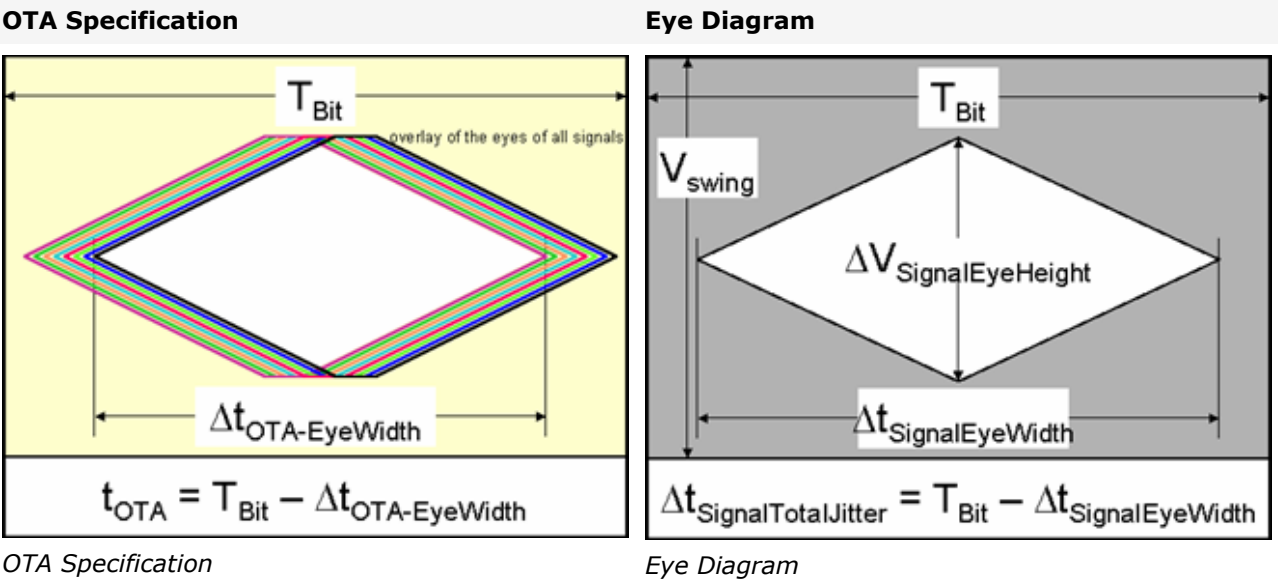
Vector period

Minimum vector period ^[9]	62.5 ps ^[10]
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^[8] edge placement range maximal -5000 ns to +15000 ns, for sequencer period > 752 ns the edge placement range is -4 to 12 sequencer periods

^[9] x16 mode for 1000ppm offset testing up to 62.43ps; 64 ps for x14; 75 ps for x12

Vector period resolution	0.001 fs
Vector Period Accuracy	±15 ppm of period setting
Minimum sequencer period	880 ps



Eye Height

Differential with $V_{swing} = 500\text{ mV}$ ^[11]

	8.0 Gbps	12.8 Gbps	E8080B/E8080E: 16 Gbps E8080C: 15.1 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	83 % of V_{swing}	83 % of V_{swing}	61 % of V_{swing}
Typical $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	80 % of V_{swing}	80 % of V_{swing}	58 % of V_{swing}
Minimum $\Delta V_{SignalEyeHeight}$ of driver+receiver pair ^[12]	72 % of V_{swing}	72 % of V_{swing}	51 % of V_{swing}

Differential with $V_{swing} = 100\text{ mV}$ ^[11]

	8.0 Gbps	12.8 Gbps	E8080B/E8080E: 16 Gbps E8080C: 15.1 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	83 % of V_{swing}	83 % of V_{swing}	55 % of V_{swing}

^[10] For E8080C: Supported up to 15.1 Gbps for all instruments except for the clock instrument.
^[11] V_{swing} is single ended swing into 50 ohm
^[12] Test system driver measured with tester receiver using PRBS 2^15 pattern, equalization is active

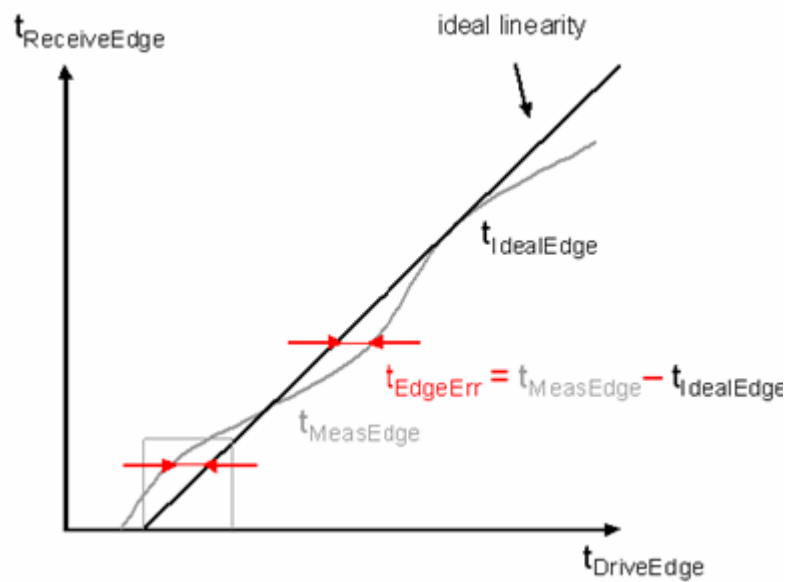
	8.0 Gbps	12.8 Gbps	E8080B/E8080E: 16 Gbps E8080C: 15.1 Gbps
Typical $\Delta V_{\text{SignalEyeHeight}}$ of driver+receiver pair	75 % of V_{swing}	75 % of V_{swing}	47 % of V_{swing}
Minimum $\Delta V_{\text{SignalEyeHeight}}$ of driver+receiver pair ^[12]	66 % of V_{swing}	66 % of V_{swing}	40 % of V_{swing}

Single Ended with $V_{\text{swing}} = 500 \text{ mV}$ ^[11]

	8.0 Gbps	E8080B/E8080E: 16 Gbps E8080C: 15.1 Gbps
Typical $\Delta V_{\text{SignalEyeHeight}}$ of driver	85 % of V_{swing}	66 % of V_{swing}
Typical $\Delta V_{\text{SignalEyeHeight}}$ of driver +receiver pair	68 % of V_{swing}	49 % of V_{swing}
Minimum $\Delta V_{\text{SignalEyeHeight}}$ of driver +receiver pair ^[11]	55 % of V_{swing}	40 % of V_{swing} ^[13]

Linearity (differential drive measured with differential receiver)

Integral Non-Linearity INL ($ t_{\text{EdgeErr}} $)	±2 ps
Differential Non-Linearity DNL	1 ps



Linearity

Clocks

Differential clock jitter at 8 GHz	≤ 900 fs rms
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^[13] pairs xxx03-04 xxx07-08 xxx11-12 xxx15-16 have minimum 35% of V_{swing}

For clocks the test system driver measured with test system receiver and measured value divided by $\sqrt{2}$. Swing is in range 100mV..500mV.

Timing modulation

Amplitude range ^[14]	± 11.5 UI
Frequency range ^[14]	1 kHz to 100 MHz

Via delay modulation with digital per pin AWG with 3072 samples.

^[14] Jitter Injection Delay Modulation AWG sampling rate depends on active timing, some limitations do apply.

PSSL Precision Timed driver

AC performance

Transition time at 1000 mV Differential Swing (20 to 80%) 35 ps $\pm$ 5 ps

DC performance (single ended V_{SE+} , V_{SE-} specified into open and maximum output current $\leq$ 50 mA)

High level range	-1.00 V to 3.51 V
Low level range	-1.01 V to 3.50 V
Swing range	20 mV to 1000 mV
Level resolution	1 mV

DC Level accuracy (single ended V_{SE+} , V_{SE-} specified into open and maximum output current $\leq$ 50 mA)

High voltage vih	± 2 mV
Differential Swing	± 2 mV $\pm$ 1%
Mid-level electrical idle	± 3 mV $\pm$ 1% of Swing
Electrical idle max voltage delta between both legs	± 2 mV

Impedance characteristic

source impedance	50 ohm $\pm$ 2.5 ohm
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PSSL Precision Timed receiver

Specification are valid as well for protocol-aware driver and Pin Scale 1600 compatible mode.

AC performance

Intrinsic transition time at 1000 mV Differential Swing (20 to 80%)	35 ps $\pm$ 5 ps
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DC performance differential

Input voltage V_{SE+} , V_{SE-} range	-1.01 V to 3.51 V
Threshold voltage V_{THDIFF} range ^[15]	-600 mV to + 600 mV
Threshold Voltage V_{THDIFF} accuracy ^[16]	± 2 mV $\pm 0.5\%$ of Threshold
Threshold voltage V_{THDIFF} resolution	± 1 mV
Common Mode Comparator range	-400 to 3400 mV
Common Mode Comparator accuracy	± 10 mV
Squelch Comparator range	25 to 120 mV
Squelch Comparator accuracy	± 10 mV

DC performance single ended

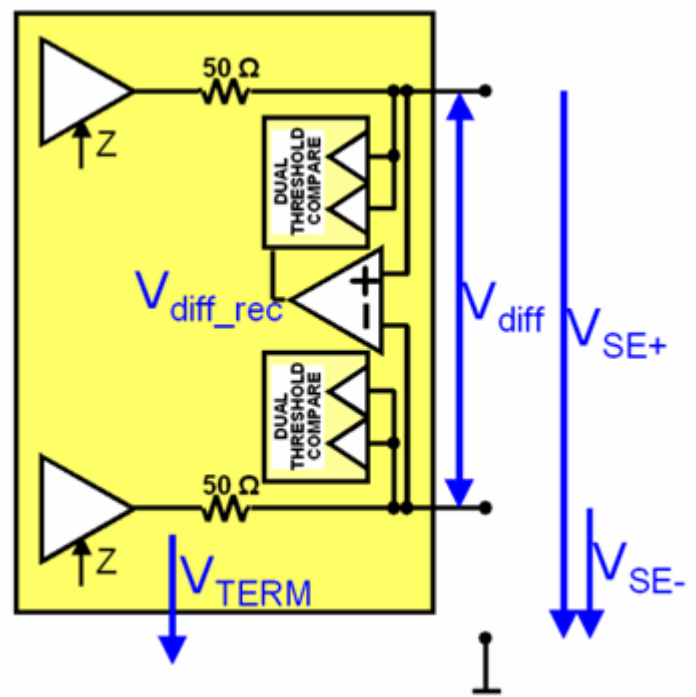
Input voltage V_{SE+} , V_{SE-} and threshold V_{TH} range	-1.01 V to 3.51 V
Threshold voltage V_{TH} resolution	± 1 mV

Termination

Single ended / center tap differential impedance	50 ohm $\pm$ 2.5 ohm
Termination level range	-1.00 V to 3.51 V
Termination Voltage accuracy	± 2 mV
Termination voltage resolution	± 1 mV

^[15] Differential threshold compares against the difference of $V_{SE+} - V_{SE-}$

^[16] With common mode voltage $V_{CM} = (V_{SE+} + V_{SE-})/2$ in range 0..1V. If V_{CM} outside 0..1V, Threshold Voltage accuracy ± 7 mV $\pm 1\%$ of Threshold Voltage



Receiver

PSSL compatible operation mode timing

OTA/EPA

Test method: TX to RX loop using PRBS 2¹⁵ pattern and verified with $V_{\text{swing}} = 200\text{mV}$ and 500mV into 50 ohm and speeds up to 1600 Mbps, loop includes standard high speed FR4 DUT board traces. For $V_{\text{swing}} = 100\text{ mV}$ Per-Board OTA < ±200 ps.

System OTA/EPA

Overall Timing Accuracy (t_{OTA}) < ±200 ps

Edge Placement Accuracy ($t_{\text{EPA}}=t_{\text{OTA}}/2$) < ±100 ps

Per-Board OTA/EPA

Overall Timing Accuracy per board (t_{OTA}) < ±160 ps

Edge Placement Accuracy per board ($t_{\text{EPA}}=t_{\text{OTA}}/2$) < ±80 ps

Edge placement

Edge placement resolution 1 ps

Edge placement range ^[17] -4 to 32 sequencer periods

Vector period

Minimum vector period ^[18] 625 ps

Vector Period Resolution 0.001 fs

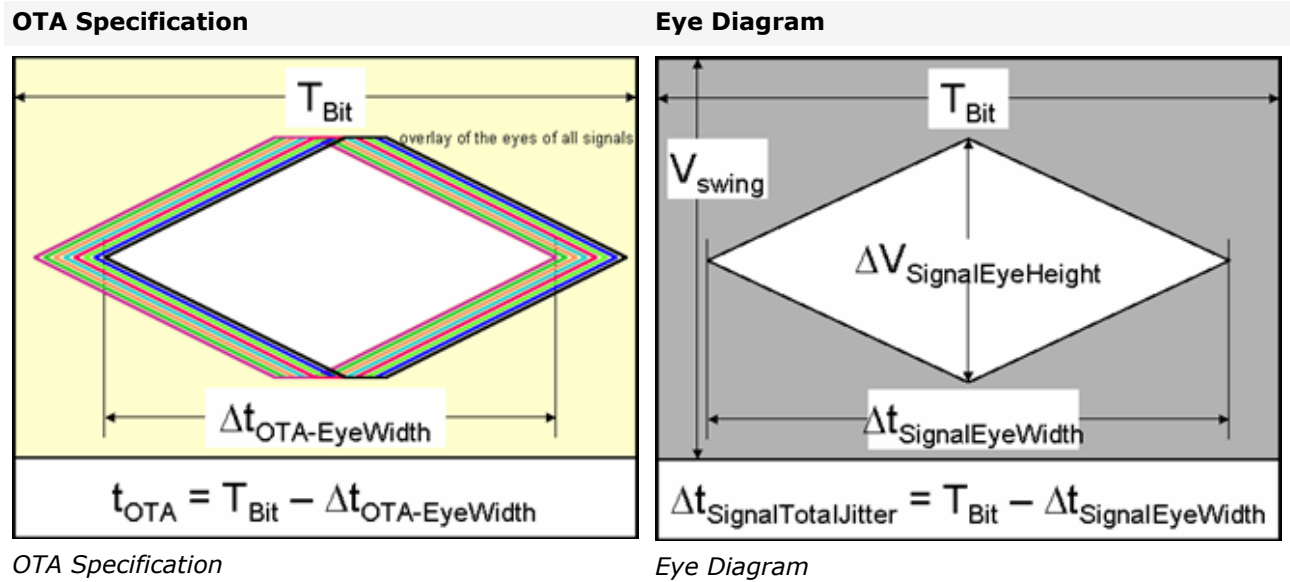
Vector Period Accuracy ±15 ppm of period setting

Minimum sequencer period 2.5 ns

Maximum sequencer period 15625 ns

^[17] edge placement range maximal -5000 ns to +15000 ns, for sequencer period > 752 ns the edge placement range is -4 to 12 sequencer periods

^[18] x4 mode for drive and receive



Eye Height

Single Ended with $V_{swing} = 500\text{ mV}$ ^[19]

	500 Mbps	1 Gbps	1.6 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	89 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	71 % of V_{swing}	68 % of V_{swing}	54 % of V_{swing}
Minimum $\Delta V_{SignalEyeHeight}$ of driver+receiver pair ^[19]	66 % of V_{swing}	29 % of V_{swing}	15 % of V_{swing}

Differential with $V_{swing} = 100\text{ mV}$ ^[20]

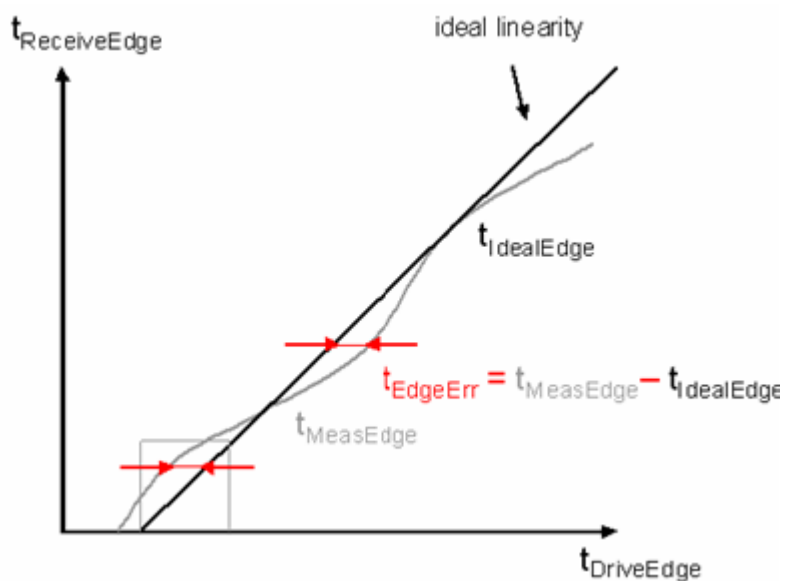
	500 Mbps	1 Gbps	1.6 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	85 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	80 % of V_{swing}	68 % of V_{swing}	35 % of V_{swing}
Minimum $\Delta V_{SignalEyeHeight}$ of driver+receiver pair ^[19]	74 % of V_{swing}	37 % of V_{swing}	11 % of V_{swing}

Linearity

Integral Non-Linearity INL ($ t_{EdgeErr} $)	$\pm 5\text{ ps}$
Differential Non-Linearity DNL	5 ps

^[19] Verified with $V_{swing} = 500\text{ mV}$ into 50 ohm
^[20] Verified with $V_{swing} = 100\text{ mV}$ into 50 ohm

Single ended driver measured with single ended receiver and differential drive measured with differential receiver.



Linearity

Clocks

Differential clock jitter at 150 MHz

typical 2.1 ps rms

Test system driver measured with test system receiver and measured value divided by $\sqrt{2}$.
Swing is in range 100mV...500mV

PSSL compatible operation mode driver

AC performance (verification condition: half the noted voltage into 50 ohm)

Maximum transition time (10 to 90%)

at 1.0 V swing	0.6 ns
at 1.8 V swing	0.7 ns
at 3.0 V swing	0.8 ns
Minimum pulse width	
at 1.0 V swing	0.7 ns
at 1.8 V swing	0.8 ns
at 3.0 V swing	0.9 ns

DC performance (high, low and 3rd level, specified into open)

Level range	-1.5 V to 6.5 V
Level resolution	1.0 mV
Level accuracy	±5 mV
Minimum swing	100 mV ^[21]
Maximum swing	8 V

Impedance characteristic

Source impedance	50 ohm ± 2.5 ohm + R ^[22]
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Z-Clamp mode

High clamp voltage range	-1.2 V to 7 V
Low clamp voltage range	-2 V to 6.25 V
Maximum voltage window	9 mV
Voltage resolution	1 mV
Voltage accuracy	±100 mV

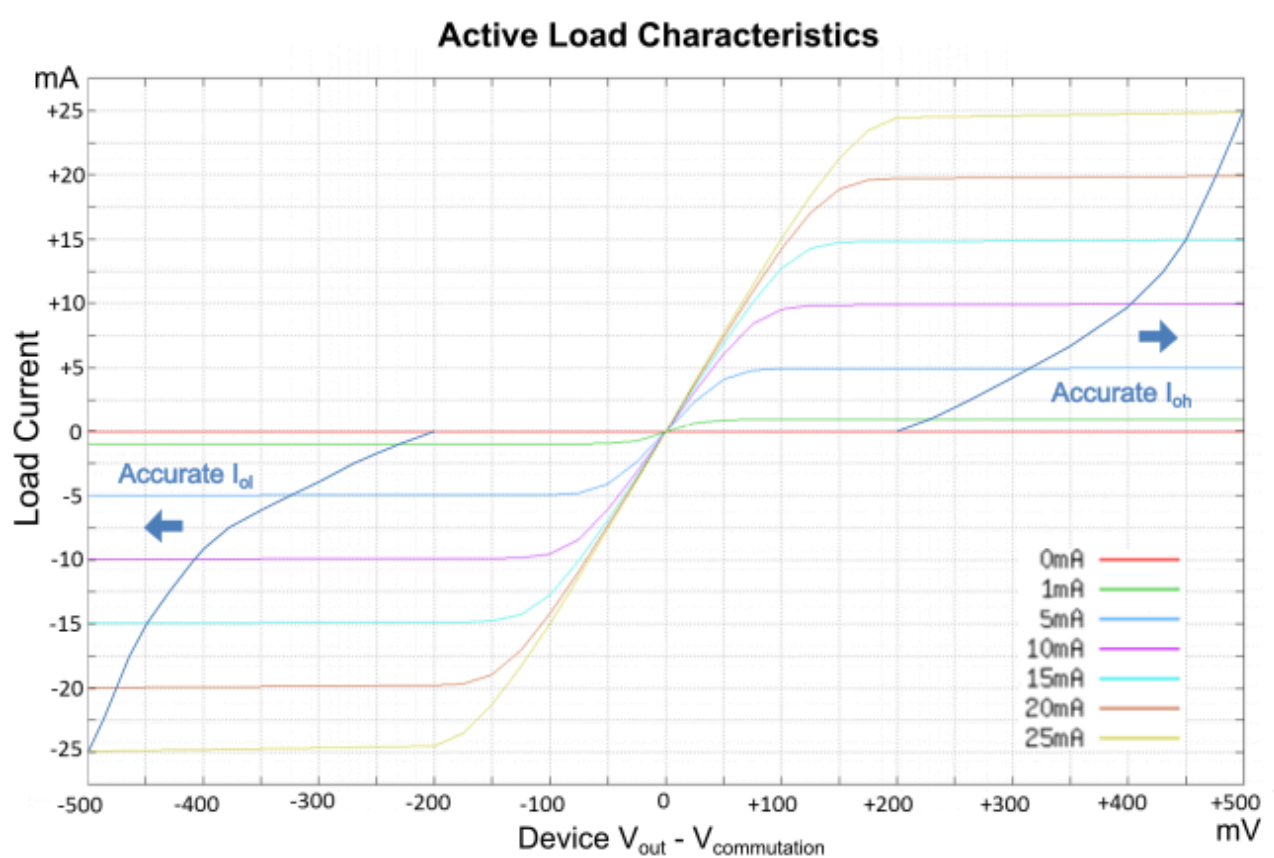
^[21] Software allows programming to 0 mV swing, below the specified minimum swing

^[22] R is the wiring resistance of ≤ 1.5 ohm

PSSL compatible operation mode programmable load

Currents (I_{oh} , I_{ol})	0 mA to 25 mA ^[23]
Current resolution	12.5 μ A
Current accuracy	$\pm 75 \mu$ A $\pm 1\%$ of max (I_{oh} , I_{ol})
Commutation voltage range (V_{com})	-1.0 V to 5.5 V ^[24]
Voltage resolution	1 mV
Voltage accuracy	± 100 mV

Typical characteristics



Active load characteristics

^[23] Sustainable total load for 32 pins per test head card may not exceed 1600 mA

^[24] -1.5 V to 6.5 V if I_{ol} , $I_{oh} < 1$ mA

PSSL compatible operation mode receiver

AC performance

Minimum detectable pulse width (single ended and differential)	600 ps
Intrinsic transition time (10 to 90%)	350 ps, typical

DC performance - single ended

Input voltage and threshold range	-1.5 V to 6.5 V
Threshold voltage resolution	±1 mV
Threshold voltage accuracy	±5 mV
Pin leakage current	±2 µA

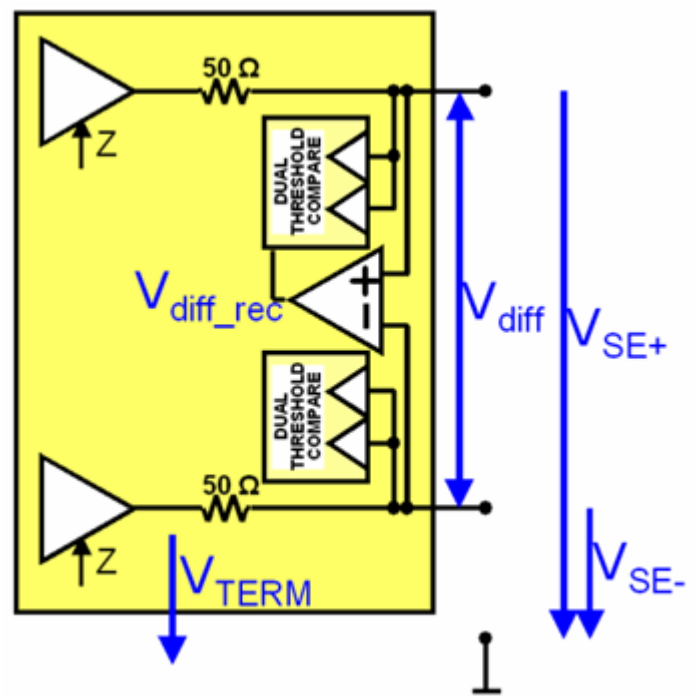
DC performance - differential

Input voltage range (V_{SE+} , V_{SE-})	-1.5 V to 6.5 V
Maximum differential input voltage (V_{diff})	8 V
Differential receive voltage range ($V_{diff_rec} = V_{SE+} - V_{SE-}$)	-1.0V to 1.0V
Threshold voltage resolution	±1 mV
Threshold voltage accuracy	±5 mV
Pin leakage current	±2 µA

Termination

Single ended / center tap differential impedance	50 ohm ± 2.5 ohm + R ^[25]
Termination level center tap range	-1.5 V to 6.5 V
Termination voltage resolution	±1 mV
Termination voltage accuracy	±5 mV
Termination voltage accuracy	±5 mV
Switchable pull-down resistor impedance	14.3 kohm ± 10%

^[25] R is the wiring resistance of ≤ 1.5 ohm



Receiver

PSSL Parametric Measurement Unit (Per-Pin PMU)

Voltage force / measure

Maximum Voltage range dependent on selected current range

Current Range	Voltage Range
within range 1 (± 40 mA)	-2.0 V to 5.75 V
within range 2..5 ($\leq \pm 1$ mA)	-2.0 V to 6.50 V

Voltage force

Voltage resolution	200 μ V
Voltage accuracy	$\pm(3 \text{ mV} + I \cdot R)$ ^[26]

Voltage measure (P/F and value) using level comparators

Voltage resolution	1 mV
Voltage accuracy	$\pm(5 \text{ mV} + I \cdot R) \pm 0.5\%$ of reading ^[26]

Voltage measure (value) using ADCs

Voltage resolution	75 μ V
Voltage accuracy in range -2.0 V to 6.5 V	± 4 mV
Voltage accuracy in range 0.0 V to 3.3 V	± 2 mV
Relative voltage accuracy in range 0.0 V to 3.3 V [27]	± 1 mV

Current force / measure

	Range	Resolution	Measure accuracy	Force accuracy
Range 1	± 40 mA	20 μ A	$\pm 50 \mu\text{A} \pm 0.5\%$ of reading	$\pm 50 \mu\text{A} \pm 0.2\%$ of setting
Range 2	± 1 mA	0.5 μ A	$\pm 1.25 \mu\text{A} \pm 0.5\%$ of reading	$\pm 5.0 \mu\text{A} \pm 0.5\%$ of setting
Range 3	$\pm 100 \mu\text{A}$	50 nA	$\pm 200 \text{ nA} \pm 0.5\%$ of reading	$\pm 500 \text{ nA} \pm 0.5\%$ of setting
Range 4	$\pm 10 \mu\text{A}$	5 nA	$\pm 50 \text{ nA} \pm 0.5\%$ of reading	$\pm 100 \text{ nA} \pm 0.5\%$ of setting
Range 5	$\pm 2 \mu\text{A}$	1 nA	$\pm 10 \text{ nA} \pm 0.5\%$ of reading	$\pm 40 \text{ nA} \pm 0.5\%$ of setting

Sustainable total load for 32 pins per test head card may not exceed 1600 mA

^[26] I is the actual current, R is the wiring resistance of ≤ 0.5 ohm

^[27] In two consecutive voltage value measurements with different force currents, the accuracy of the calculated delta of the forced currents

Range 1 force current up to $\pm 56\text{ mA}$ for voltages $\leq 3.3\text{ V}$. Range 1 force accuracy applies.

Current force for relative measurement

Relative accuracy ^[27]	$\pm 20\text{ }\mu\text{A} \pm 0.3\%$ of setting in range 1
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Voltage clamp (available in current force)

Voltage range	-2.0 V to 6.5 V
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Voltage resolution	1 mV
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Voltage accuracy	$\pm 100\text{ mV}$
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PSSL Time Measurement Unit (Per-Pin TMU)

This topic covers time measurement unit performance data.

For details on the operation of the TMU, refer to "Theory of operation of the Time Measurement Unit (TMU) (Topic 254558)".

DC performance

Specifications of receiver apply (see *PSSL compatible operation mode receiver* on page 19 and *PSSL Precision Timed receiver* on page 12).

AC performance

Maximum input frequency	16 Gbps
Linearity error	±5 ps
Accuracy, single timestamp to ARM (t0)	±25 ps
Accuracy, skew/propagation delay (between different pin resources) ^[28]	2 x (t _{EPA} +15 ps)
Intrinsic jitter (RMS)	3 ps
Resolution	1 ps

Other

Maximum sampling rate ^[29]	50 Msps
Measurement mode	Single ended & differential

^[28] t_{EPA} of Pin Scale 1600 compatible operation mode
^[29] The maximum number of samples is limited by the available channel memory.